

Reading the Moon from Apollo Thin Section 10020

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1 Introduction

This lunar rock 10020 was brought back by the Apollo 11 astronauts from Mare Tranquilitatis, one of the Moon's dark basaltic plains. When you look up at the Moon, you can clearly distinguish two contrasting regions: the dark maria and the bright highlands. The maria appear as dark patches, which are actually vast plains of solidified basaltic lava; the highlands are the lighter regions that surround the maria, representing the Moon's older crust.

A sequence of PPL and XPL images of sample 10020 at 5° intervals was acquired from Virtual Microscope [1]. These images were subsequently aligned to ensure a consistent field of view during stage rotation.

2 Petrographic Observations

2.1 Central bright grains

In PPL:

- Appearance: Appearing bright /white; no pleochroism.
- Relief: High relief. Grain boundaries appear sharp and stand out.
- Shape (habit): Grains look irregular.
- Texture: Fractured, with irregular cracks
- Other features: They show irregular internal boundaries consistent with adjoining grains. They are surrounded by finer groundmass. They locally contain small dark grains and small clear /cloudy grains.

In XPL:

- Inference colors: Relatively high-order interference colors (vivid colors that shift as the sample is rotated)
- Extinction: Straight extinction (parallel to crystallographic axes)

Interpretation:

The bright centerpiece is best interpreted as a clustered olivine phenocryst aggregate, consisting of multiple subgrains with similar optical orientations. Olivine grains locally contain inclusions

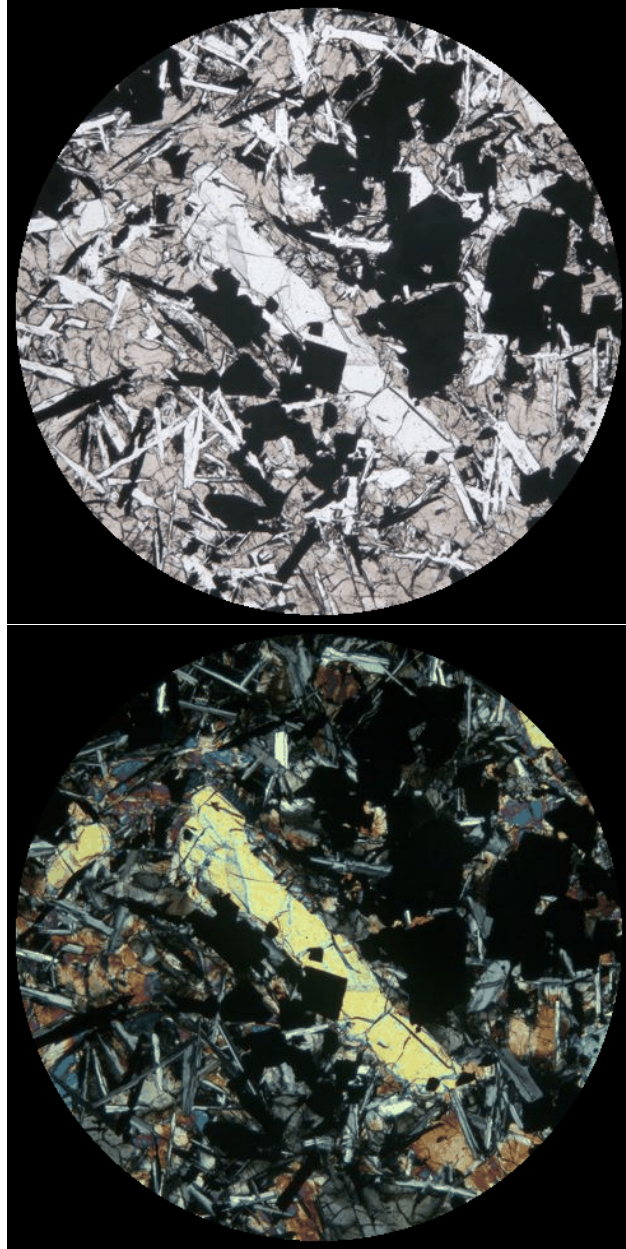


Figure 1: Sample 10020-r1 in PPL (top) and XPL (bottom). Field of view 2.0 mm

of (1) opaque minerals, likely trapped during crystal growth; and (2) likely melt inclusions.

2.2 Pale brown grains

In PPL:

- Appearance: Pale brown (groundmass), subdued appearance
- Relief: Moderate to high relief
- Shape: Blocky
- Cleavage: straight, parallel lines within the grains
- Other features: They are distributed throughout the groundmass. Some occur along the margins of the centerpiece.

In XPL:

- Interference colors: Moderate interference colors
- Extinction: Uniform extinction

Interpretation:

The pale brown grains scattered throughout are likely pyroxene.

2.3 Lath-shaped crystals

In PPL:

- Appearance: Bright
- Relief: Low relief (harder to see boundaries). They look less crisp.
- Shape: Elongated laths
- Other features: They form a network of laths throughout the groundmass. They appear less distinct than higher-relief minerals. They are intergrown with pale brown grains.

In XPL:

- Interference colors: Low-order interference colors (gray, white) when not in extinction.
- Polysynthetic twinning: They appear as alternating light and dark stripes (most diagnostic feature).
- Extinction: They go extinct in segments. During rotation, one set of stripes goes dark (extinction), and the adjacent stripes are still bright, and then they switch; which creates a striped flashing pattern.

Interpretation:

The lath-shaped crystals are plagioclase.

2.4 Completely black grains in all frames

1. Appearance: Complete black in both PPL and XPL at every rotation angle
2. Other features: They are distributed throughout and also occur as inclusions within the central grains. They occupy /fill the spaces between earlier-formed minerals. They are

relatively large, abundant, and range from acicular forms to blocky forms.

Interpretation:

These are opaques (likely ilmenite or Fe-Ti oxides in lunar samples). The relatively coarse and abundant oxide minerals (likely ilmenite) occur between earlier-formed silicate grains, suggesting crystallization from a Fe-Ti-rich melt during the later stages of cooling. The inclusions are likely Cr-spinel (chromite).

2.5 Small irregular bright grains

- Appearing bright /white in PPL.
- Irregular shapes
- Relatively high interference colors in XPL
- Not dominant in this specific field

Interpretation:

These are possibly minor olivine.

2.6 Melt inclusions

- Appearing tiny, rounded or irregular inside a host mineral
- Clear or cloudy in PPL and black /mixed in XPL

Interpretation:

These are likely melt inclusions, tiny trapped droplets of ancient melt inside centerpiece olivine.

2.7 Glasses

Appearing white in PPL but always black in XPL

2.8 Overall texture

This thin section of sample 10020 shows olivine as relatively large grains, plagioclase as lath-shaped crystals forming a framework, pyroxene occurring between plagioclase and along olivine margins, and oxides present between silicate grains. These observed relationships are consistent with a **porphyritic basalt texture** in which early-formed olivine is enclosed within a finer-grained groundmass.

The arrangement of plagioclase laths and pyroxene suggests an **intergranular** to locally **subophitic texture**. The occurrence of pyroxene along olivine margins indicates later crystallization from the remaining melt, while the presence of relatively coarse oxide minerals reflects crystallization from a Fe-Ti-rich melt during the later stages of cooling. Together, these features record a sequential crystallization history during progressive cooling of a basaltic melt.

Petrographic observations provide a first-order identification of minerals based on optical properties; however, definitive mineralogy is established through chemical and crystallographic analyses [2, 3]. Based on optical characteristics and textural relationships, the minerals in this thin

section of sample 10020 are interpreted as olivine, pyroxene, plagioclase, and Fe-Ti oxides. These identifications are consistent with previous chemical and microanalytical studies, which confirm this mineral assemblage [4]. Each mineral carries information about the conditions under which it formed, and their collective arrangement reveals the history of the rock. In sample 10020, the textural relationships between olivine, pyroxene, plagioclase, and oxide minerals allow reconstruction of a crystallization sequence and cooling history. These interpretations extend beyond mineral identification, providing insight into the processes of magma evolution and, ultimately, the thermal evolution of the Moon.

3 Segmentation of Opaques

You may want to give it a try to segment opaque phases in the PPL and XPL images, allowing the texture of the sample to be visualized more clearly: <https://huggingface.co/spaces/pingwang2026/segmentation-of-opaques-in-petrographic-images>

The Moon may seem still and cool, but it was once thermally active, and alive with molten rock flowing across its surface and shaping its geology.

References

- [1] "10020 Ilmenite basalt," *The Virtual Microscope*, The Open University. [Online]. Available: <https://www.virtualmicroscope.org/content/10020-ilmenite-basalt>
- [2] W. D. Nesse, *Introduction to Optical Mineralogy*, 4th ed. New York, NY, USA: Oxford University Press, 2014.
- [3] W. A. Deer, R. A. Howie, and J. Zussman, *An Introduction to the Rock-Forming Minerals*, 3rd ed. London, U.K.: Mineralogical Society, 2013.
- [4] C. Meyer, *Lunar Sample Compendium*. Houston, TX, USA: NASA Johnson Space Center, 2011. [Online]. Available: <https://curator.jsc.nasa.gov/lunar/lsc/>